Effect of Bag Carrying Method on Upper Trapezius Muscle Symmetry During Walking

A. Mereminsky, A. Kiperman and B. Mazri

School of Medical Engineering, Afeka – Tel Aviv Academic College of Engineering, Israel

Lecturers: Dr. Naftali Sara & Prof. Anat Ratnovsky

1. Introduction

- Daily bag carrying alters mechanical loading on the upper body and spine.
- Asymmetric load carriage increases muscular effort and may contribute to fatigue or musculoskeletal strain.
- Surface EMG of the upper trapezius allows quantitative assessment of muscle activation and symmetry under different carrying conditions.

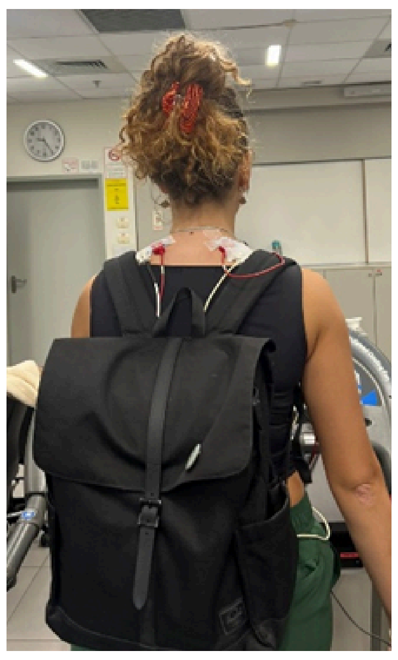
2. Objective

To examine how different bag-carrying methods, relative to walking without a bag, affect upper trapezius muscle activation and left - right symmetry during walking.

3. Methods

- Experimental Conditions: Walking on a treadmill (4.6 km/h, no Incline) under three conditions: no bag (baseline), backpack (two straps), and shoulder bag carried on the right side. **Load was set to 10% of body weight.**
- Data Acquisition: Surface EMG signals were recorded bilaterally from the upper trapezius muscles using a BIOPAC system.

Figure 1. Backpack Condition



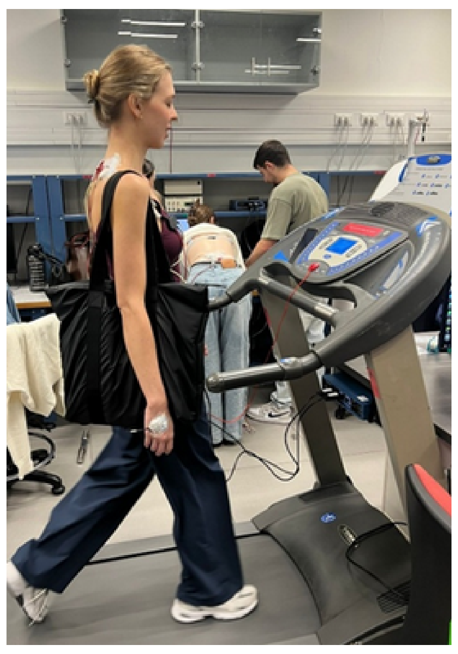


Figure 2. Shoulder bag condition

- Signal Processing: Raw EMG signals were first demeaned to remove DC offset, band-pass filtered (20-450 Hz), and converted to RMS values using consecutive 30-second windows.
- Outcome Measure: Muscle symmetry was quantified using an **Asymmetry Index (AI)** comparing right and left muscle activation.

$$Asymmetry\ Index = \frac{|right\ RMS - left\ RMS|}{(right\ RMS + left\ RMS)}$$

4. Results

4.1 Baseline:

Natural left - right asymmetry was observed during unloaded walking, with a group-averaged AI of 35% across subjects (N=3).

Asymmetry Index percentage
7%
54.3%
44.1%

Figure 3. Asymmetry index Across Participants

4. Results cont.

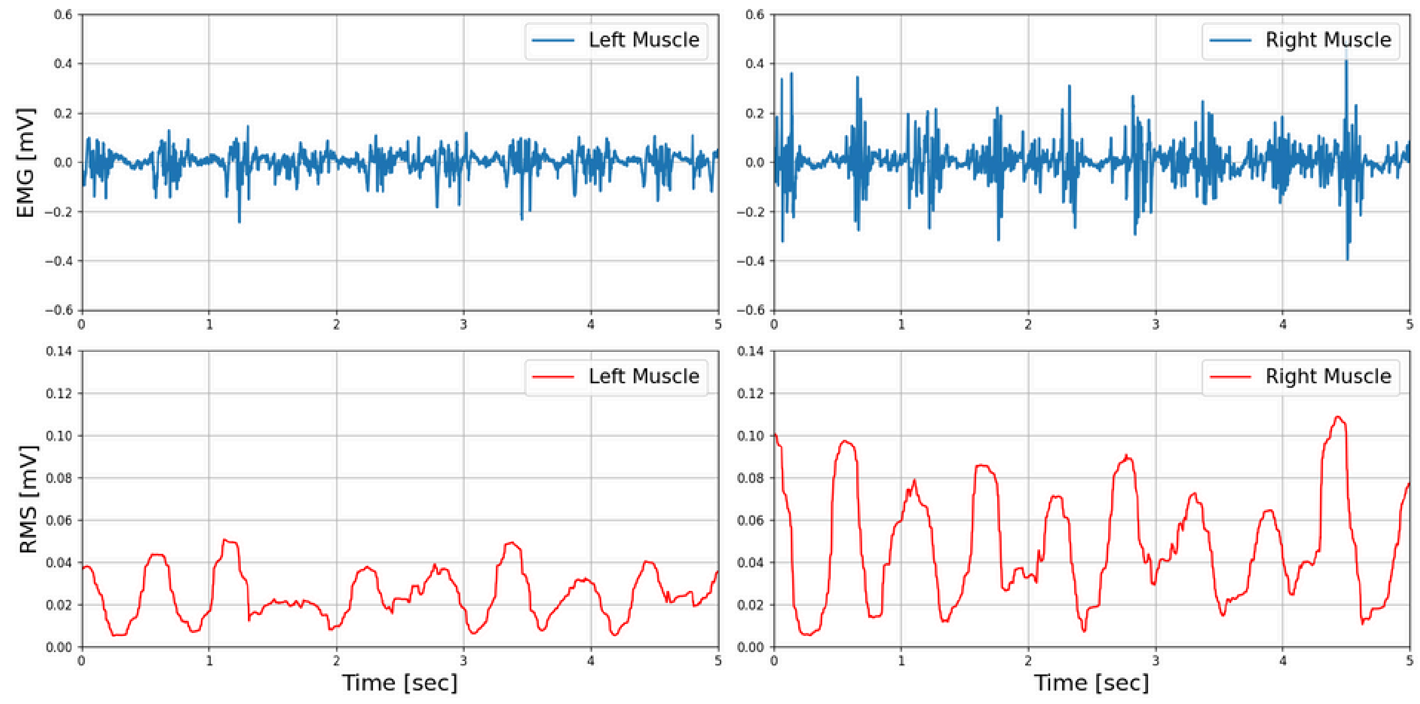


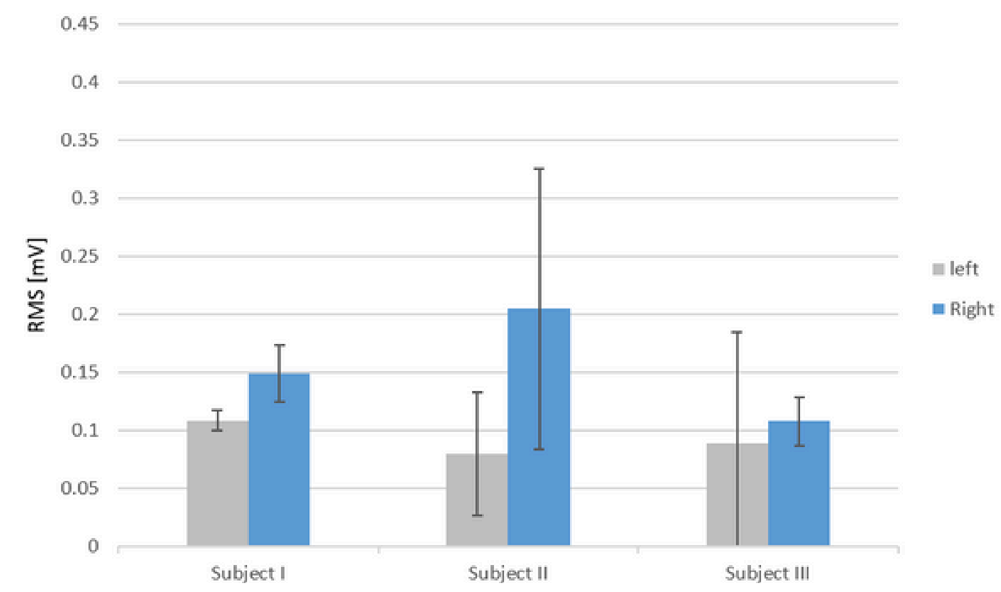
Figure 4. Raw EMG Signal and Processed RMS

Figure 4 shows a representative participant with right side dominant activity during baseline walking.

4.2 Backpack:

group-averaged AI **decreased by 35%,** indicating improved muscle symmetry under symmetric load carriage, compared to baseline.

Figure 5. Left vs. Right RMS Across Participants – Backpack Condition



4.3 Shoulder Bag:

group-averaged AI increased by 25%, reflecting increased muscle asymmetry under uneven load distribution.

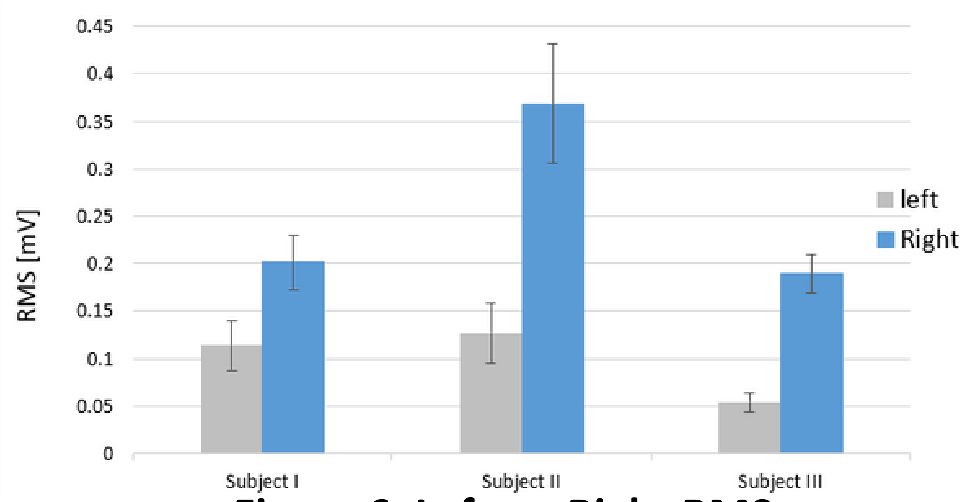


Figure 6. Left vs. Right RMS Across Participants – Shoulder Bag Condition

4.4 Overall Comparison

Mean RMS values varied with the Asymmetry Index, with overall higher right side muscle activity averaged across subjects in all conditions.

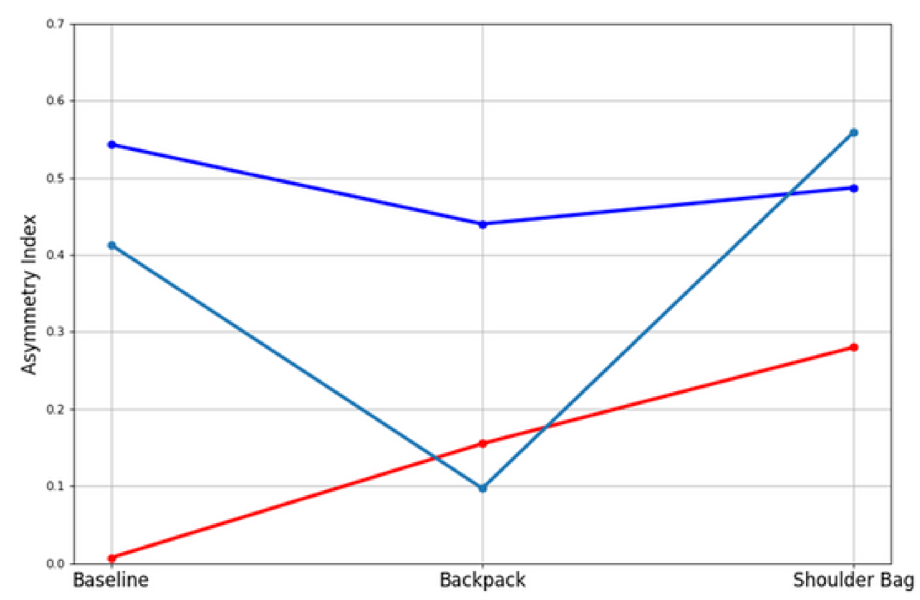


Figure 7. Asymmetry Index Across Conditions for Each Subject

5. Summary & Conclusion

- Shoulder bag carriage increased asymmetry and right side RMS activity across participants, while **backpack carriage promoted more symmetric activation** in some cases, compared to baseline.
- Inter subject variability was observed, including notable variability in the baseline condition, suggesting that **individual movement patterns may influence asymmetry measures,** Given the small sample size (N=3), **the findings are exploratory.**